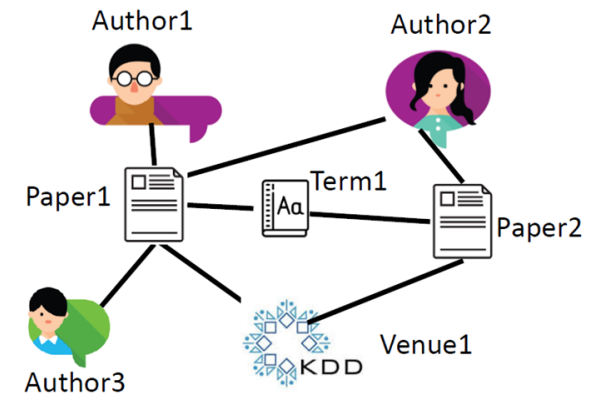
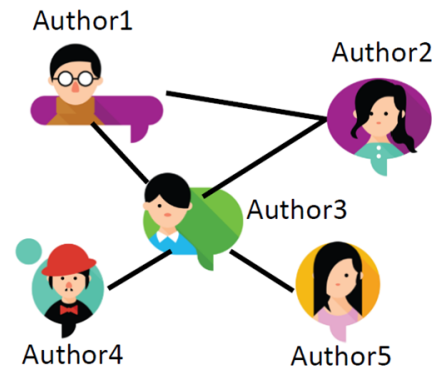


Mining Rich Data Types

- ❑ Mining Text Data
- ❑ Mining Spatial-Temporal Data
- ❑ Mining Graph and Networks 

Mining Graph and Networks

- ❑ Graph/Networks: consist of (1) a set of nodes and (2) a collection of edges, to model complex systems, e.g., online social platform
- ❑ Categorization
 - ❑ attributed vs. plain graphs:
node/edge attributes exist or not.
 - ❑ dynamic vs. static graphs:
graph topology/attribute change or not.
 - ❑ homogeneous vs. heterogeneous graphs:
nodes/edges are of the same type or not.



Homogeneous vs. heterogeneous networks

Mining Graph and Networks: Problems

- ❑ Graph Modeling: to study and simulate the generation mechanisms.
 - ❑ Motivation: (1) understanding how graphs are created and formed, (2) helping protect user privacy by anonymizing social networks.
 - ❑ Graph models:
 - ❑ Erdős–Rényi model: assumes certain probability distributions.
 - ❑ realistic graph generators: mimic one or more properties of real-world graphs, e.g., preferential attachment based (Barabási–Albert model).
 - ❑ deep generative models: mimic the real graphs with neural networks.
 - ❑ Challenge: heterogeneous networks

Mining Graph and Networks: Problems

- ❑ Heterogeneous Network Mining: capturing rich semantics
 - ❑ Core concept: **meta-path** represents a sequence of relations, e.g., author-paper-author $\Rightarrow$ co-authorship relation.
 - ❑ Applications of meta-path
 - ❑ Similarity search: improving ranking accuracy
 - ❑ Node classification: similarity matrices can be used to infer node labels
 - ❑ Clustering: representing different semantics
 - ❑ heterogeneous network embedding: constructing neighborhood
 - ❑ Challenge: automatically inferring structure and semantics from input data

Mining Graph and Networks: Problems

□ Knowledge Graph Mining

- Knowledge graph (KG): a directed heterogeneous graph to link concepts and entities through human-interpretable semantics, e.g., DBPedia, Google KG.
- KG construction methods: labor intensive, relying on pre-specified ontology
 - Extractive construction: information extraction, e.g., semantic labeling.
 - Language model-based
 - Representation learning-based: link prediction, entity resolution
- Challenges: (1) automatic knowledge graph construction and dynamic update, (2) medical AI, conversational AI, and academic search engine